

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

A) Create a file by using

```
shivam@MSI123:~$ nano shell.c
```

Then press enter and the code terminal will open then write a code

```
shivam@MSI123: ~  
GNU nano 7.2 shell.c  
#include <stdio.h>  
#include <string.h>  
  
int main() {  
    char input[100];  
  
    while (1) {  
        printf("myShell> ");  
        fgets(input, sizeof(input), stdin);  
  
        // Remove newline  
        input[strcspn(input, "\n")] = 0;  
  
        // Exit condition  
        if (strcmp(input, "exit") == 0) {  
            printf("Exiting Shell...\n");  
            break;  
        }  
  
        printf("You entered: %s\n", input);  
    }  
  
    return 0;  
}
```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command

```
shivam@MSI123: ~  
shivam@MSI123:~$ nano shell.c  
shivam@MSI123:~$ gcc shell.c -o shell
```

After compiling to get the output enter command

```
shivam@MSI123: ~  
shivam@MSI123:~$ nano shell.c  
shivam@MSI123:~$ gcc shell.c -o shell  
shivam@MSI123:~$ ./shell  
myShell> HI  
You entered: HI  
myShell> I am learing OSSP  
You entered: I am learing OSSP  
myShell> This is my firt programming  
You entered: This is my firt programming  
myShell> |
```

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

open()

syntax: open(" filename.txt", modes, permissions)

Example: open("First_Prgm.txt", O_CREAT | O_WRONLY, 0644);

Table 1: Modes of Open system call

Flag	Purpose
O_RDONLY	Open for read only
O_WRONLY	Open for write only
O_RDWR	Open for both read and write
O_CREAT	Create file if it does not exist
O_TRUNC	Truncate (empty) an existing file
O_APPEND	Append data to the end of the file
O_EXCL	Fail if file already exists (used with O_CREAT)

read()

Syntax: read(int fd, void *buf, size_t count);

Example: read(fd, buffer, 100)

#include <stdio.h>

#include <unistd.h>

int main()

```
{
    char buffer[100];
    int n;
    printf("Enter some text: ");
    n = read(0, buffer, sizeof(buffer) - 1);
    buffer[n] = '\0'; // Null-terminate the string
    printf("You entered: %s", buffer);
    return 0;
}
```

write()

Syntax: write(int fd, const void *buf, size_t count);

Example: write(fd, message, 11);

#include <unistd.h>

int main()

```
{
    write(1, "Hello World\n", 12);
    return 0;
}
```

```
shivam@MSI123: ~$ nano File1.tx|
```

```
GNU nano 7.2 File1.tx *
Operating Systems Lab
Using open(), read(), write(), and close() system calls.
|
```

```
shivam@MSI123: ~  
shivam@MSI123:~$ nano File1.tx  
shivam@MSI123:~$ nano copy.c  
shivam@MSI123:~$ |
```

```
GNU nano 7.2 copy.c  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <stdlib.h>  
  
int main()  
{  
    int source, destination;  
    char buffer[1024];  
    ssize_t bytesRead;  
  
    // Open File1.txt for reading  
    source = open("File1.txt", O_RDONLY);  
  
    if (source < 0)  
    {  
        printf("Error opening File1.txt\n");  
        return 1;  
    }  
  
    // Open File2.txt for writing (create if it doesn't exist)  
    destination = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);  
  
    if (destination < 0)  
    {  
        printf("Error opening File2.txt\n");  
    }  
}
```

[Read 44 lines]

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo	M-/
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo	M-?

```
shivam@MSI123: ~$ nano File1.tx
shivam@MSI123: ~$ nano copy.c
shivam@MSI123: ~$ gcc copy.c -o copy|
```

```
shivam@MSI123: ~$ nano File1.tx
shivam@MSI123: ~$ nano copy.c
shivam@MSI123: ~$ gcc copy.c -o copy
shivam@MSI123: ~$ ./copy|
```

```
shivam@MSI123: ~  
shivam@MSI123:~$ nano File1.tx  
shivam@MSI123:~$ nano copy.c  
shivam@MSI123:~$ gcc copy.c -o copy  
shivam@MSI123:~$ ./copy  
Content copied successfully.  
shivam@MSI123:~$ cat File1.txt|
```

```
shivam@MSI123: ~  
shivam@MSI123:~$ nano File1.tx  
shivam@MSI123:~$ nano copy.c  
shivam@MSI123:~$ gcc copy.c -o copy  
shivam@MSI123:~$ ./copy  
Content copied successfully.  
shivam@MSI123:~$ cat File1.txt  
Operating Systems Lab  
Using open(), read(), write(), and close() system calls.  
shivam@MSI123:~$ cat File2.txt  
Operating Systems Lab  
Using open(), read(), write(), and close() system calls.  
shivam@MSI123:~$ |
```